ANTICIPY HARDWARE

Parts list and CAD brief

One-sentence product definition

Anticipy is a small, wearable ambient-audio pendant that records conversations, sends audio to an iPhone over Bluetooth, stores missed audio when the phone is away, and gives simple feedback through one button, one light and a quiet vibration.

1. What the pendant does

1 HEAR	2 SHRINK	3 SEND / SAVE	4 CONFIRM
Microphones capture ambient speech.	The processor compresses the audio.	Bluetooth streams to iPhone; flash holds any backlog.	Button, light and haptic give simple status feedback.

Functional target

- 16 kHz mono speech capture; compressed to roughly 16 kbps with silence suppression.
- Live Bluetooth Low Energy connection to the Anticipy iPhone app when the phone is nearby.
- At least 20 hours of offline audio capacity; oldest unsent files are backfilled when the phone returns.
- Target approximately 16 hours between charges; final runtime must be proven on the physical build.
- One multifunction button, discreet RGB status light, haptic confirmation and USB-C charging.
- No screen, camera, speaker, cellular radio or Wi-Fi in this version.

2. Industrial-design direction

Area	Direction
Shape	Compact rounded pebble / soft pill. Not a long recorder-shaped capsule.
Final target envelope	Approximately 36 × 34 × 12 mm. Treat this as the CAD goal; validate it against imported component STEP models.
No-PCB working envelope	Approximately 38 × 35 × 19 mm because complete breakout boards and hand wiring need more thickness.
Outside material	Natural satin-silver aluminum faces over a structural polycarbonate body and RF window.
Branding	Small, centered ANTICIPY laser mark. No large logo, graphics or visible fasteners.
Details	One discreet microphone opening, one flush side button, one tiny light window and a small jewelry bail at the top seam.

Material note

PLAUD publicly specifies aluminum alloy plus polycarbonate, but not its proprietary alloy, resin or finish recipe. Anticipy should use the same material families and match the pale silver appearance with a physical colour/finish sample; it should not claim an identical private formula.

3. Route A — working mock-up without a custom PCB

Use this route to make the fastest functional engineering sample. It can perform the core jobs, but it will be thicker and more handmade inside.

Qty	Job	Exact part	CAD envelope	Why it is there
1	Brain + Bluetooth + mic	Seeed Studio XIAO nRF52840 Sense — SKU 102010469	21 × 17.8 mm	Processor, BLE 5, PDM mic, IMU, USB-C and LiPo charging
1	Offline storage board	Adafruit MicroSD SPI/SDIO breakout — PID 4682	25.4 × 22.8 × 3.5 mm	Stores audio when the phone is away
1	Memory card	Kingston Industrial microSD 8 GB — SDCIT2/8GB	15 × 11 × 1 mm	Removable storage; far more than the 20-hour target
1	Battery	Adafruit protected LiPo 3.7 V 500 mAh — PID 1578	29 × 36 × 4.75 mm	Power; reserve extra room for its lead and protection tab
1	Prototype haptic	Adafruit vibrating mini motor disc — PID 1201	Ø10 × 2.7 mm	Simple vibration confirmation
1 ea.	Motor switch parts	AO3400A N-MOSFET + 1N4148W diode + 100 Ω / 100 kΩ resistors	Small hand-wired parts	Lets the XIAO control the motor safely
1	Button	C&K KMR221GLFS tactile switch	4.2 × 2.8 × 1.9 mm	Single multifunction control
1	Light pipe	Clear 1 mm PMMA light pipe or molded PC feature	Ø1–1.5 mm	Carries the XIAO RGB light to the exterior
1 set	Internal assembly	34 AWG silicone wire, Kapton tape, Poron foam, thin VHB	Allow 1–2 mm routing space	Insulation, strain relief and anti-rattle support

Route A placement

- Battery forms the lower layer; XIAO and microSD sit above it in opposite corners.
- Keep the XIAO antenna at the polycarbonate edge. No aluminum, battery or wire may sit directly over the antenna end.
- Put the microphone directly behind the acoustic opening with a small gasket so sound cannot leak around the port.
- Use pockets and foam compression so no board, battery or motor can move or rattle.
- This route is for a working prototype. Do not force it into the 12 mm final shell by crushing the battery or bending boards.

Simple verdict

No custom PCB = faster and easier, but approximately 19 mm thick. It is suitable for proving the function and for a realistic appearance mock-up, not the final thin production interior.

4. Route B — compact final-looking version with a custom PCB

Use one purpose-built 4-layer board, approximately 30 × 28 × 0.8 mm, to reach the thin compact shape. This is the correct route for a polished investor or production-intent unit.

Qty	Job	Preferred part	CAD envelope	Placement note
1	Bluetooth processor module	Raytac MDBT50Q-1MV2 (nRF52840)	10.5 × 15.5 × 2.05 mm	Pre-certified BLE module; place at polycarbonate RF edge
2	Digital microphones	TDK InvenSense T5838	3.50 × 2.65 × 0.98 mm each	Dual-mic speech capture; use separate acoustic ports and gaskets
1	Offline flash	Winbond W25N04KV, 4 Gb QSPI NAND	Select 6 × 8 mm or 5 × 5 mm package	512 MB raw capacity; sufficient for 20 hours at target bitrate with margin
1	Power + charging	Nordic nPM1300 PMIC	Use manufacturer land pattern	LiPo charging, regulated rails, fuel gauge and battery protection inputs
1	Haptic driver	Texas Instruments DRV2605L	2.0 × 2.0 mm WCSP option	Clean, controlled haptic effects
1	Haptic actuator	Vybronics VG0832013D LRA	Ø8 × 3.25 mm	Thin wearable-grade vibration with adhesive backing
1	Button	C&K KMR221GLFS	4.2 × 2.8 × 1.9 mm	Flush multifunction control
1	Status light	1.5 mm side-fire RGB LED + molded light pipe	Reserve 2 × 2 × 2 mm	Subtle status only
1	Charging/data	Low-profile mid-mount USB-C receptacle	Reserve 9 × 4 × 7 mm around port	Charging, firmware flashing and service access
1	Battery	Protected 3.7 V LiPo, 400–500 mAh	Reserve 30 × 20 × 6 mm	Final supplier/part number must be locked from a certified cell drawing
As req.	PCB support parts	Crystals, passives, ESD, USB-C CC resistors, test pads, antenna keep-out	Electrical-layout dependent	Electrical engineer completes these from the reference designs

Custom PCB rules

- Use a 4-layer, 0.8 mm PCB with rounded corners. Import the exact manufacturer STEP models before freezing the enclosure.
- The Bluetooth antenna must face the polycarbonate radio window and obey Raytac's exact copper, battery and metal keep-out drawing.
- Put both bottom-port microphones over PCB acoustic holes; seal each to the enclosure with a tiny closed-cell gasket and acoustic mesh.
- Keep the battery on the opposite side from the antenna and give it a protected pocket with no sharp edges, screw tips or compression.
- Expose SWD programming/test pads, battery test points and microphone test access before sealing the cosmetic faces.

Simple verdict

Custom PCB = more engineering, but it removes the stacked breakout boards and is the only sensible way to reach approximately 36 × 34 × 12 mm with a clean, repeatable interior.

5. CAD construction brief

Part	Material / finish	CAD requirement
Front and rear faces	0.60–0.65 mm 5052-H32 aluminum; fine unidirectional brush; natural clear anodize	Softly crowned faces, narrow hairline seam, no exposed screws. Laser-mark ANTICIPY after finishing.
Middle frame / RF window	True polycarbonate, colour-matched pale silver or warm light grey	Structural internal pockets plus a non-metal antenna path. Avoid brittle cosmetic resin for final handled units.
Internal carrier	Black or neutral PC/PA12	Positive pockets for PCB, cell, haptic and light pipe; 0.2–0.4 mm foam allowance; no loose parts.
Microphone ports	Black acoustic mesh behind shell	One small port per microphone, typically 0.5–0.8 mm; keep ports away from clothing rub surfaces.
Button	Matching anodized aluminum or plated PC cap	Flush or 0.1 mm proud; positive click; prevent accidental presses.
Pendant attachment	Small stainless jewelry bail	Attach at the top seam. Do not cut a large chain hole through the battery/electronics volume.
Service opening	Hidden bottom USB-C opening	Cable must insert fully without touching the aluminum edge; add internal strain support.

Required CAD deliverables

- Fully assembled STEP model and an exploded STEP model.
- Separate manufacturing files for aluminum faces, polycarbonate body, internal carrier, button and light pipe.
- 2D drawing with overall dimensions, wall thicknesses, seam targets, microphone/USB/button locations and material callouts.
- One cross-section showing the complete internal stack and all clearances.
- STL/3MF files for prototype printing plus neutral STEP files for machining/quoting.
- A simple photoreal render in pale satin silver, front / back / side / worn-on-chain views.

6. Acceptance checklist

Gate	Pass condition
Appearance	Looks like a small piece of jewelry; no visible wires, screw heads, print lines, glue squeeze-out or crooked branding.
Fit	Nothing rattles; button clicks; USB cable inserts; microphones and light pipe align; enclosure closes without forcing the battery.
Radio	Bluetooth connects through the final metal/polycarbonate enclosure, not only with the case open.
Audio	Both speech and quiet-room audio are intelligible; clothing rub and haptic vibration do not overwhelm the microphone.
Offline	Audio saves with the phone absent and backfills in order after reconnection.
Power	Charges safely, reports battery level, stays cool and meets the measured runtime target.
Safety	Protected cell, insulated edges, strain relief, charging-temperature test and no battery compression.

What is locked vs. what still needs engineering

Locked: product behavior, compact pebble direction, aluminum + polycarbonate construction, controls and the two hardware routes. Still to be finalized by the electrical engineer: the certified production battery SKU, complete passive component BOM, schematic, PCB routing, antenna verification, charging limits and physical runtime/audio tests.

Reference links

Seeed XIAO nRF52840 Sense: https://wiki.seeedstudio.com/XIAO_BLE/

PLAUD published material / size comparison: https://support.plaud.ai/hc/en-us/articles/53770668740761-What-is-the-difference-between-Plaud-NotePin-and-NotePin-S

Raytac MDBT50Q-1MV2: https://www.raytac.com/product/ins.php?index_id=24

Nordic nRF52840: https://www.nordicsemi.com/Products/nRF52840

TDK T5838 microphone: https://invensense.tdk.com/en-us/products/microphone/t5838

TI DRV2605L: https://www.ti.com/product/DRV2605L

Vybronics VG0832013D: https://www.vybronics.com/coin-vibration-motors/lra/v-g0832013d